Application of RFID and Computer Vision for the Inventory Management System

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Abstract. The RFID technology can be used for item tracking and inventory control. However, the problem such as miss reading and ghost reading usually occur in RFID implementation and has impact on low accuracy of inventory management system. In this study, the computer vision is used to solve the problem of miss reading and ghost reading in RFID. The RFID and computer vision can act as ears and eyes respectively, thus by combining both technologies; it is expected to increase the accuracy of system. The result of experiment has showed that the combination of RFID and computer vision has increased the system accuracy, as the computer vision can help the RFID system to detect the miss reading and ghost reading.

Keywords: RFID · Miss reading · Ghost reading · Computer vision

1 Introduction

RFID (Radio Frequency Identifier) technologies promise to revolutionize future inventory management. They are used to replace the barcode system such that the ID that identifies an object can be accessed wirelessly over a distance [1]. The RFID applications hold tremendous promise, such as RFID tag monitoring. Tag monitoring is a key component in applications like item tracking and inventory control. By frequently scanning its inventory, a retailer can quickly determine if anything is missing, and act accordingly. In addition, the RFID technology has been implemented successfully in Supply Chain Management to reduce the inventory losses [2], for identification and monitoring of agricultural animals [3, 4], food traceability system [5], and healthcare applications to improve patient care as well as reduce overall costs [6].

An RFID passive tag is attached to each item to be monitored. An RFID reader is periodically dispatched to collect all the tags information. A passive RFID tag has limited processing abilities, no power supply, and can only communicate when an RFID reader queries the tag. Since a passive RFID tag is much cheaper than an active tag, passive tags are likely to be deployed on commodity items such as clothing and other retail goods. An active tag is likely to be used on more expensive products.

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However, there are at least two problems exist in RFID tag monitoring, such as miss reading and ghost reading. In miss reading, the reader is not able to read the tag in some certain position. This may be due to many reasons such as bad reading angles, blocked signals, or signal dead zones [7]. As opposite, the ghost reading or simply ghost read, is the receipt by the reader of data that is interpreted as valid but was not communicated by a single tag, i.e. the reader receives incorrect data which it interprets as valid data. The identifier received as a result of a ghost read may never have been issued by the numbering authority, may be a valid identifier but not exist on any tag received at the facility experiencing that ghost read, or may exist at that facility but is not in an area where it could possibly be read by the reader experiencing the ghost read [8].

Furthermore, the automatic face analysis which includes, e.g., face detection, face recognition, and facial expression recognition has become a very active topic in computer vision research. Local Binary Pattern (LBP) is one of the most-widely used approach, mainly in face description [9]. In addition, the combination of computer vision (image processing) and RFID technology has been implemented in attendance management system [10], and proved that these two technologies can be implemented together. Thus in this study, the computer vision application and RFID technology will be utilized to solve the miss reading and ghost reading of tag.

2 Methodology

In this study, the application of RFID system in automotive industry is considered as case study. The main function of RFID application in automotive industry is to track/trace the spare part of car. By utilizing this functionality, the certain location of products can be presented in detail from the whole supply chain. For the case of car manufacturing, the spare part of car are placed into special boxes, and special RFID label and tags are attached in boxes. The special RFID tag is presented as the rectangle with the full color, as can be seen in the Fig. 1. In the Fig. 1, the staffs are checking the RFID tag on the box (which the car spare part are placed inside). The color of tag is different with box and presented as bright/contrast color. By designing the tag into

Fig. 1. Special RFID tag, which can easily recognized by computer vision system.

special shape and color, the pattern of tags can be generalized/trained, thus it can be detected by computer vision technique in the last step.

As can be seen in the Fig. 2, the proposed tracking system should recognize the total number of products which are placed on the trolley/cart. The reader is placed nearby the cart, as well as the camera. Both of the RFID reader and camera are connected to the computer. The reader captures the tags which are attached on the product. The total number of tags is counted and presented in the system (computer). In addition, the camera captures every images and send it to system to be recognized by computer vision technique, whether in the image there are products or not (recognized based on the tag pattern and its color). In the last step, the total number of tags from reader are compared with the total number of tags from images, if the result is same, the program/system will show the correct decision, otherwise wrong decision. This computer vision technology is used to double check the tracking system of RFID technology.

In this study, the computer vision technology utilizes LBP (Local Binary Pattern) to recognize the pattern of tags. The LBP (Local Binary Pattern) operator [9] is one of the best performing texture descriptors and it has been widely used in various applications. It has proven to be highly discriminative and its key advantages, namely, its invariance to monotonic gray-level changes and computational efficiency, make it suitable for demanding image analysis tasks. The LBP operator was originally designed for texture description. The operator assigns a label to every pixel of an image by thresholding the 3×3-neighborhood of each pixel with the center pixel value and considering the result as a binary number. Then, the histogram of the labels can be used as a texture descriptor.

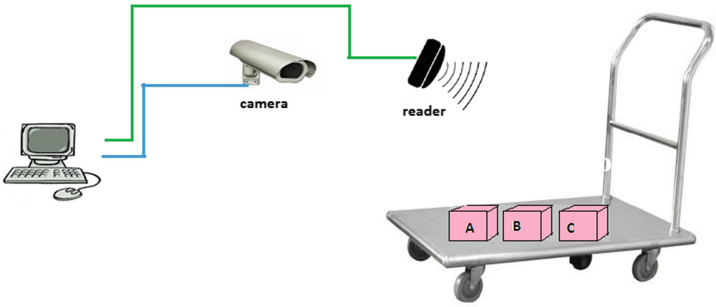

Fig. 2. Integrated RFID and camera for product tracking.

3 Result and Discussion

In this study, the prototype of application is written in Java programming language, and utilize the *openCV* as library which the LBP is included. The *ALR-9900* RFID Reader is used for demonstration while the two IP cameras, *SNV-7084R Samsung* are used to capture the images of the products. The computer *Core i3* is used in this experiment, to receive the data from reader as well as images from IP cameras. The system is

successfully implemented and correctly recognized the number of products which are placed in front of the reader and cameras.

As can be seen in Fig. 3, the boxes which have the special colored tag are placed on the cart or trolley. The total boxes in this experiment are 16 and placed it into two rows, 8 boxes in the each side. For each side, the IP camera is placed around 3 m from the boxes and facing directly to the boxes. The RFID antenna reader is placed near by the products, so it can easily read the tags. As can be seen in Fig. 3, the Graphical User Interface of the system shows two cameras are capturing of the products in each row and the RFID reader is reading the tags of the products. The system calculates the total tags which are read by the reader and by the cameras. The decision of the system is *right*, which mean the number of products which are detected from computer vision technique and RFID tracking system is same/correct.

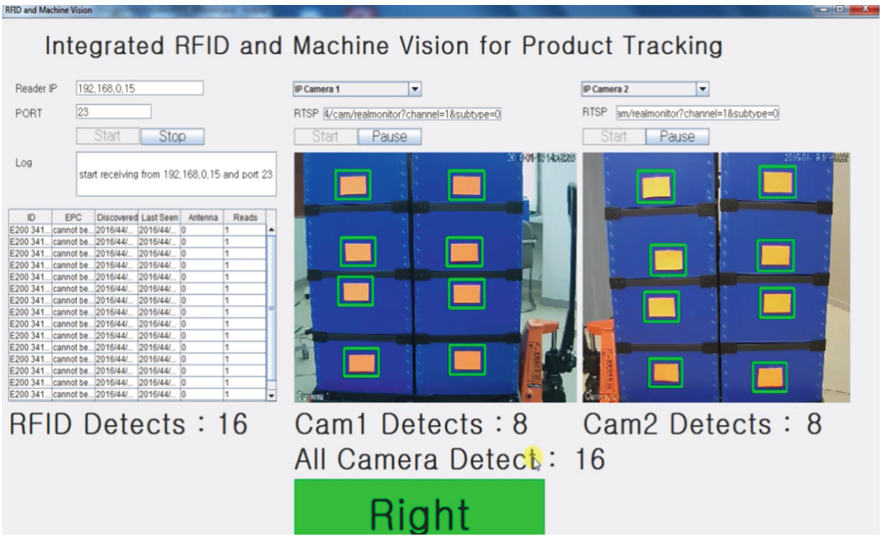

Fig. 3. The RFID reader and computer vision successfully detect number of products.

Furthermore, radio frequencies in reading areas might cause RFID readers to obtain inexistent RFID objects. Thus, the system should be able to avoid this problem. As can be seen from Fig. 4, the boxes are moving on the trolley or cart, while the cameras and reader are capturing the images and tags respectively. In this scenario, the reader is detecting 17 tags, while the computer vision technology is correctly detecting 16 products. The final decision of system is *wrong*, which is the result from RFID reader and computer vision are not match. The RFID reader has showed *wrong* decision, as ghost reading has been occurred in this scenario. In here, we can conclude the computer vision technology is able to avoid the ghost reading.

In addition, RFID readings are noisy in actual situations. This may be due to many reasons such as bad reading angles, blocked signals, or signal dead zones. Miss readings

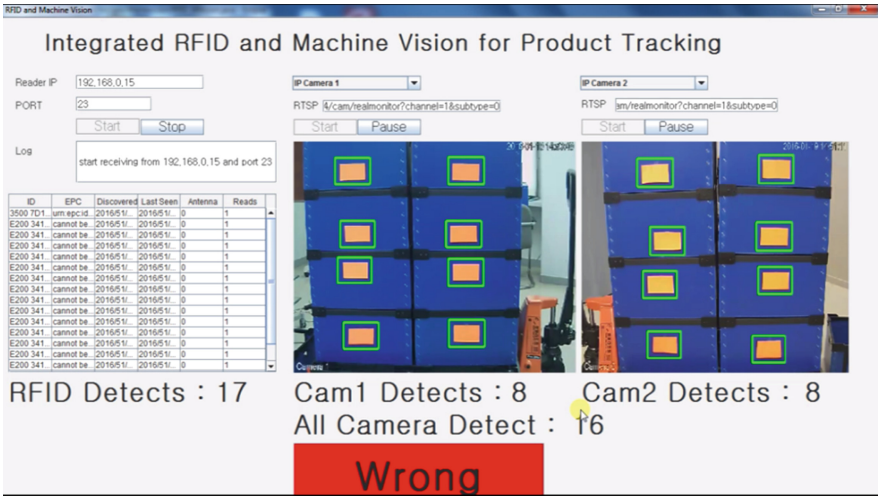

Fig. 4. The different result is detected which is caused by ghost reading.

result in lack of objects' certain positions. As can be seen in Fig. 5, the 16 boxes are moving on the trolley or cart, and reader and cameras are capturing the tag and images respectively at the same time. In this scenario, the reader is capturing 15 tags, while the computer vision technology is correctly detecting 16 boxes which are gathered from two cameras. Thus the final decision is *wrong*, the result from both systems are not correct. In this case, the miss reading has been occurred on RFID reader, while the computer vision system can show real number of products. Thus this computer vision technology is able to avoid the problem of missing tag.

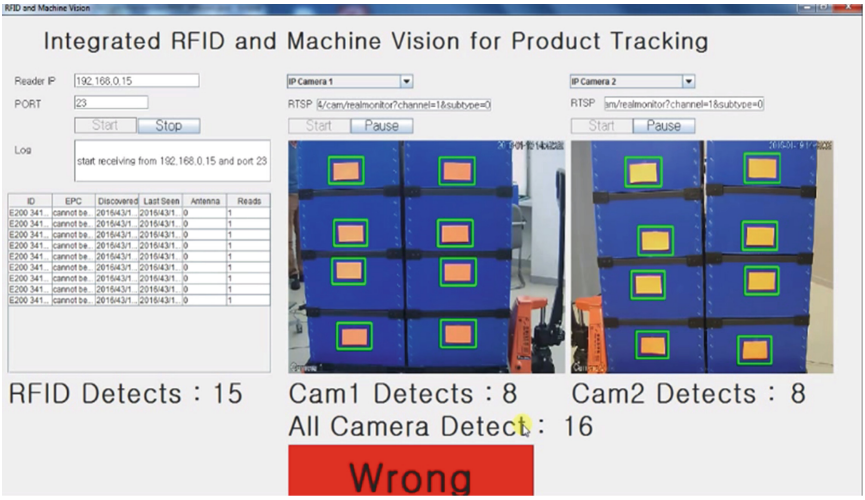

Fig. 5. The different result is detected which is caused by miss reading.

4 Conclusion

In this study, the development of RFID tracking system and computer vision system are presented. The both system are combined together for better decision making, to avoid miss reading and ghost reading which usually occurs in RFID system. Both systems, RFID tracking system and computer vision are developed on Java programming language. The case study on car manufacturing is presented, and prototype of system is developed. The test lab is presented to simulate the real case of car manufacturing. The scenario which involves the products move on the cart is considered. The result of simulation/test lab has showed good result that the computer vision technology can increase the accuracy of system by avoiding miss reading and ghost reading problem which usually occurs in RFID. The computer vision technology can be used to help the RFID tracking system to double check the total number of products, thus it will increase the accuracy of inventory management system.

Future work could be, increasing the number of dataset for training to recognize different object/tag, different angle of images, increasing the camera distance for better decision making and cost analysis of the system.

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